

Tribhuvan University  
Faculty of Humanities & Social Sciences  
OFFICE OF THE DEAN

2022

## Bachelor in Computer Applications

Course Title: **Computer Graphics and Animation**

Code No: **CACS305** Semester: **V**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

**Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

### Group B

2. Define horizontal and vertical retrace of electron beam. Explain architecture of random scan display system.
3. How does decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center has (10, 5) using mid-point circle algorithm.
4. What do you mean by window to viewport transformation? Explain 2D viewing pipeline.
5. Generate the points between the end points of a line viz. (2, 2) and (9, 6) by using DDA line drawing algorithm.
6. Define visible surface detection. Outline the Z-buffer algorithm along with its advantages and disadvantages.
7. What do you mean by Virtual Reality? Explain the types of Virtual Reality.
8. Define reflection transformation and derive the 2D reflection matrix along x-axis and y-axis in homogeneous coordinate.

### Group C

*Attempt any TWO questions[2x10=20]*

9. What is the need of homogeneous coordinate system in geometric transformation system? Find the new co-ordinate of rectangle ABCD whose center is at (4, 2) is reduced to half of its size and center will remain same. The co-ordinate of ABCD are A(0, 0), B(0, 4), C(8, 4) and D(8, 0).
10. Explain Sutherland-Hodgeman Polygon Clipping algorithm and trace it to clip a polygon with end points P(10, 30), Q(25, 35), R(35, 50), S(60, 30), T(45, 28), U(40, 15) against a window whose lower left corner is at (15, 25) and upper right corner is at (50, 40).
11. What do you understand by object space and image space method? Explain Flood Filled algorithm with its applicability and concept.